

Matrices and linear combinations have geometric meaning

Ex: $\{\text{arrows on board}\} \cong \mathbb{R}^2$ ↖ isomorphic

to work in coordinates we need bases

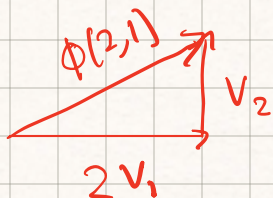
$\mathbb{R}^2$ has a basis $[(1,0), (0,1)]$

basis for this:

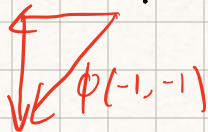


$\{\text{arrows}\}$ has basis $[v_1, v_2]$

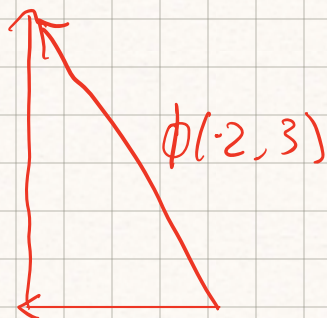
$$\phi(2,1) = \phi(2(1,0) + (0,1)) = 2\phi(1,0) + \phi(0,1) = 2v_1 + v_2$$



$$\phi(-1,-1) = -v_1 - v_2$$

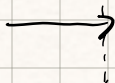


$$\phi(-2,3) = -2v_1 + 3v_2$$

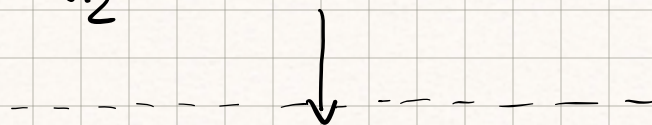


$\{\phi(1,x) \mid x \in \mathbb{R}\}$ where are the ends of these arrows? - depends on x , but along a line $\perp$ to v_1

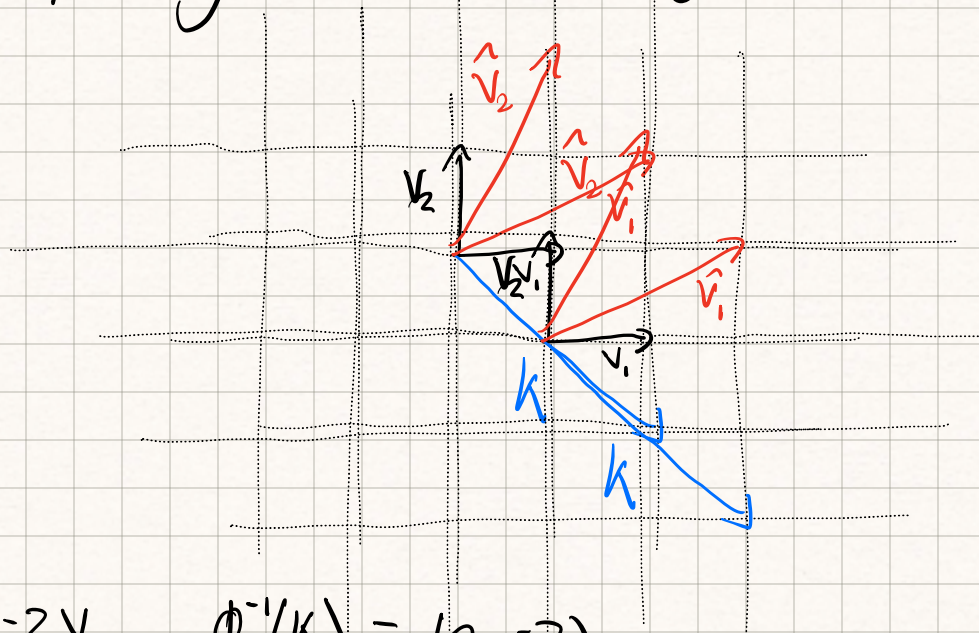
$$\phi(1,x) = v_1 + xv_2$$



$$\phi(x, -1) = x v_1 - v_2$$



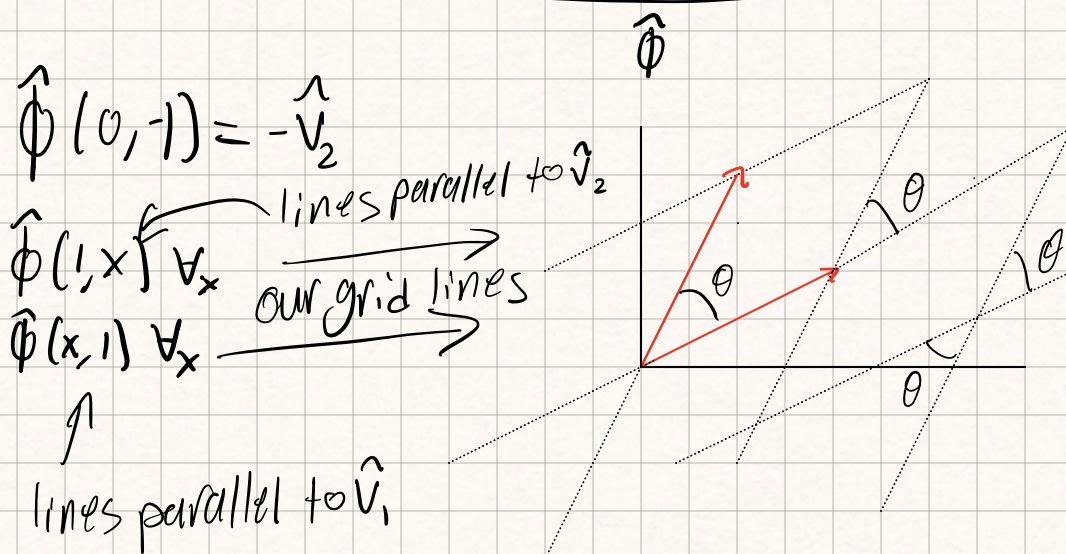
if we keep doing this: we get grid lines (our matrix)



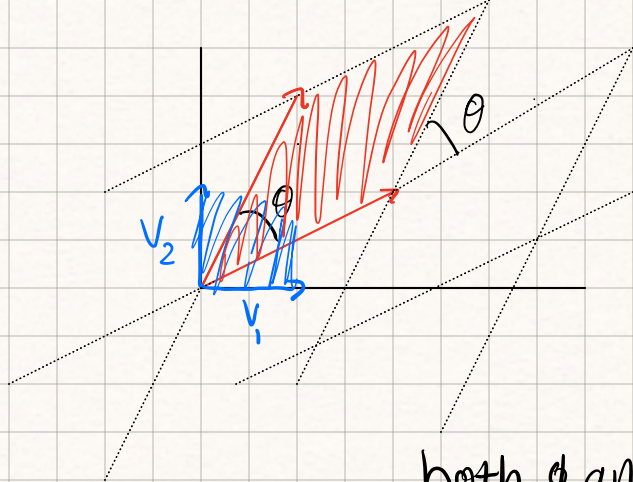
$$k = 2v_1 - 2v_2 \quad \phi^{-1}(k) = (2, -2)$$

What if we picked a diff basis?

$\mathbb{R}^2$ has same basis, {arrows} has basis $[\hat{v}_1, \hat{v}_2]$



so we get 2 sets of coords for the plane 1 from $[v_1, v_2]$ and one from $[\hat{v}_1, \hat{v}_2]$



$$\phi(\{(a,b) \mid 0 \leq a, b \leq 1\})$$

$$\hat{\phi}(\{(a,b) \mid 0 \leq a, b \leq 1\})$$

both areas are non 0
 both ϕ and $\hat{\phi}$ are invertible funcs!